

Software Requirements Specification

for

Coffee POS

Version 1.0 approved

Prepared by POS

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1. Introduction

1.1 Product Scope

The Coffee POS system is a web-based Point of Sale (POS) application designed to support daily operations of a coffee shop. The system enables staff (baristas) and administrators (owners/managers) to efficiently manage orders, payments, inventory, reporting, and staff accounts within a centralized platform

The primary objectives of the system are:

- To streamline the order creation and payment process.*
- To provide real-time visibility of active orders.*
- To maintain accurate inventory tracking.*
- To generate reliable sales reports for business decision-making.*
- To enforce role-based access control for operational security.*
- To ensure secure payment and refund processing.*

1.2 Document Conventions

This Software Requirements Specification (SRS) follows standard IEEE-style requirement structuring and is organized by functional and non-functional requirements derived from defined use cases.

Conventions used in this document include:

- Use Case format for describing functional requirements.*
- Requirements are written in clear, testable statements.*
- Preconditions must be satisfied before execution of a use case.*
- Postconditions define the system state after successful completion.*
- Error conditions define expected failure handling behavior.*

- *Non-functional requirements are listed under each use case and consolidated later in Section 5.*
- *System actions are described from the system perspective.*
- *Actor actions are described from the user perspective.*

1.3 Intended Audience and Reading Suggestions

Developer: To understand system functionality, constraints, and technical expectations before implementation.

Sequence for reading the document:

- *Section 1 (Introduction)*
- *Section 2 (Overall Description)*
- *Section 4 (System Features / Use Cases)*
- *Section 5 (Non-functional Requirements)*

Project manager: To review system scope, dependencies, and operational constraints.

Sequence for reading the document:

- *Section 1 (Introduction)*
- *Section 2 (Overall Description)*
- *Section 5 (Non-functional Requirements)*

Testers: To derive test cases from use cases, error conditions, and performance requirements.

Sequence for reading the document:

- *Section 4 (System Features / Use Cases)*
- *Section 5 (Non-functional Requirements)*

Owner: To understand business capabilities, reporting functions, and administrative controls.

Sequence for reading the document:

- *Section 1 (Introduction)*
- *Section 2 (Overall Description)*
- *Section 4 (System Features / Use Cases)*

1.4 References

The following documents and standards support or influence this SRS:

- *IEEE 830 / IEEE 29148 Software Requirements Specification Standard*
- *Role-Based Access Control (RBAC) security model principles*
- *Payment Card Industry Data Security Standard (PCI-DSS) guidelines for secure payment handling*
- *RESTful API design conventions*
- *Material Design UI guidelines (for tablet optimization standards)*

2. Overall Description

2.1 Product Perspective

The coffee POS system is a centralized, web-based point-of-sale solution designed specifically for coffee shop operations. It integrates order management, payment processing, inventory control, reporting, and staff management into a single streamlined platform. From an end-user perspective:

- **Baristas** can quickly create and customize orders, process payments, update order statuses, and view active order queues in real time.
- **Owners and managers** can manage menu items, monitor inventory levels, generate sales reports, export financial data, manage staff accounts, and process refunds securely.

The system replaces traditional cash registers and manual tracking systems with a modern digital interface optimized for tablet devices. It improves operational efficiency, reduces human error, ensures secure transactions, and provides actionable business insights through real-time reporting.

2.2 Operating Environment

The system includes:

- Secure Staff Login and Authentication
- Order Creation and Customization
- Order Modification and Status Updates
- Real-Time Active Orders Queue
- Payment Processing (Cash, Card, Mobile Payment)
- Refund Processing
- Menu Management (Add/Edit/Deactivate Items)
- Inventory Management with Low-Stock Tracking
- Sales Reporting (Daily, Weekly, Monthly)
- Report Export (CSV/PDF)
- Staff Account Management with RBAC
- Transaction Logging and Audit Trail

2.3 Design and Implementation Constraints

- Cloud-based web application
- Accessible via modern web browsers on tablet, desktop, and mobile devices
- Optimized for touchscreen tablet POS systems
- Backend services hosted on a secure cloud server (e.g., Linux-based server)
- Relational SQL database for persistent data storage

- Integration with PCI-DSS-compliant third-party payment gateway
- Secure HTTPS communication with TLS encryption

2.4 User Documentation

The Coffee POS system will include the following user documentation components to support staff and administrative users during deployment and operation:

1. Staff (Barista) User Guide

A digital user manual will be provided for staff users. The guide will include step-by-step instructions for:

- Logging in and logging out
- Creating and customizing orders
- Modifying orders prior to payment
- Processing cash and electronic payments
- Updating order status
- Viewing active orders queue

The Documentation will emphasize fast workflow execution and touchscreen interaction, consistent with the system's tablet-optimized design.

2. Administrator (Owner/Manager) Guide

A separate administrative guide will provide instructions for:

- Managing menu items (add, edit, deactivate)
- Managing inventory levels and thresholds
- Viewing and exporting sales reports
- Managing staff accounts and roles

- Processing refunds
- Reviewing transaction logs

Administrative documentation will include security and audit trail explanations aligned with the system's RBAC and logging requirements.

3. Quick Start Guide

A concise onboarding document will be provided to assist in initial system deployment. This will include:

- Device requirements
- Login setup procedures
- Basic order processing workflow
- Payment processing overview

4. System Messages & Notifications

The system will provide contextual help through:

- Error notifications
- Payment failure alerts
- Inventory shortage warnings
- Role-based access restriction messages

All documentation will be delivered digitally in PDF format and accessible through the Admin Dashboard.

2.5 Assumptions and Dependencies

The following assumptions and external dependencies may affect the development and operation of the Coffee POS system.

Assumptions

1. Stable Internet Connectivity

The system assumes that POS client devices maintain a stable internet connection in order to communicate with the backend server, process payments, and synchronize system data.

2. Cloud-Based Backend Availability

The system assumes backend services are hosted on a reliable cloud server environment capable of supporting concurrent POS users during normal business operations.

3. Role-Based User Accounts

The system assumes that staff and administrator user accounts are properly created and maintained so that role-based access control (RBAC) can be enforced throughout the system.

4. Valid Menu and Inventory Data

The system assumes that administrators maintain accurate menu items and inventory records so that staff users can create and process orders correctly.

Dependencies

1. Payment Gateway Services

Electronic payment functionality depends on integration with a third-party payment gateway that supports secure card or mobile payment transactions.

2. Relational Database System

The system depends on a relational SQL database to store persistent system data including orders, menu items, inventory records, user accounts, and transaction logs.

3. POS Client Devices

The system depends on POS client devices (tablet or desktop terminals) that provide network connectivity and touchscreen interaction for staff users.

4. Secure HTTPS Communication

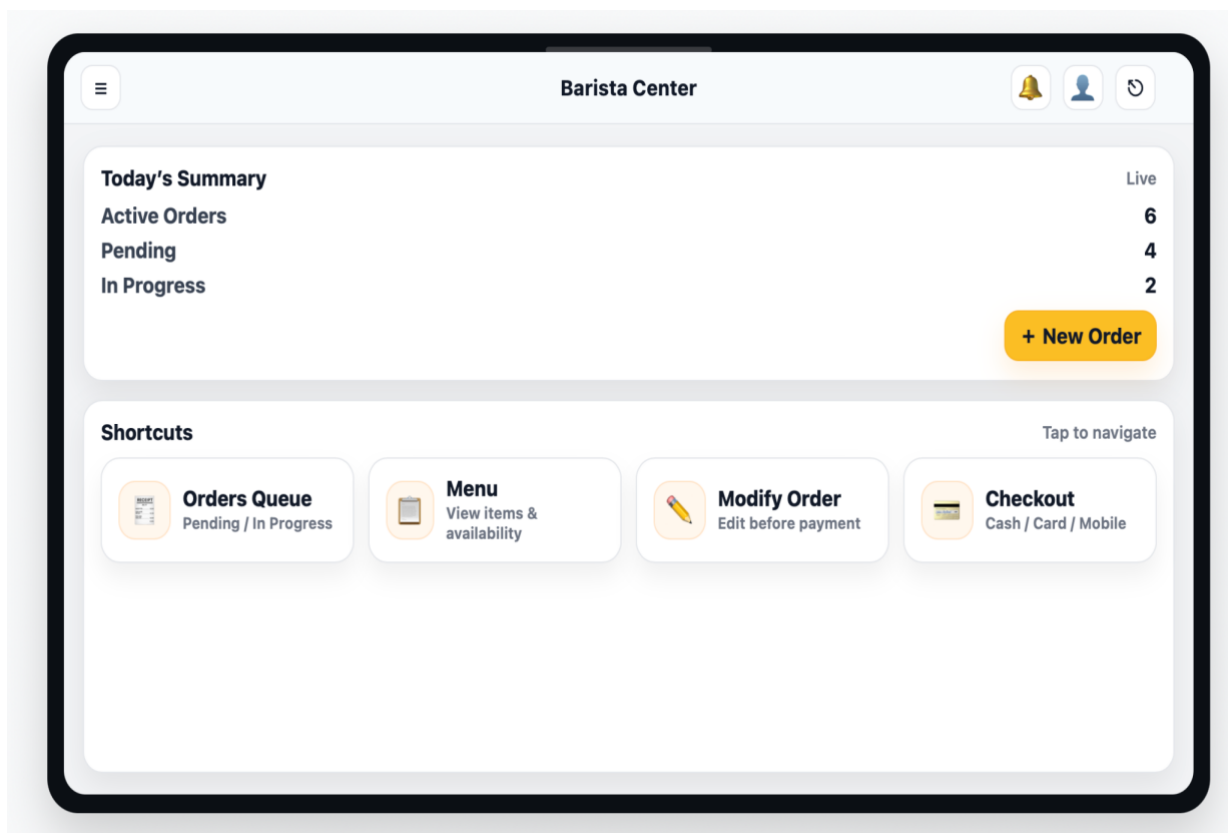
All client-server communication depends on secure HTTPS connections with TLS encryption to protect authentication data and transaction information.

3. External Interface Requirements

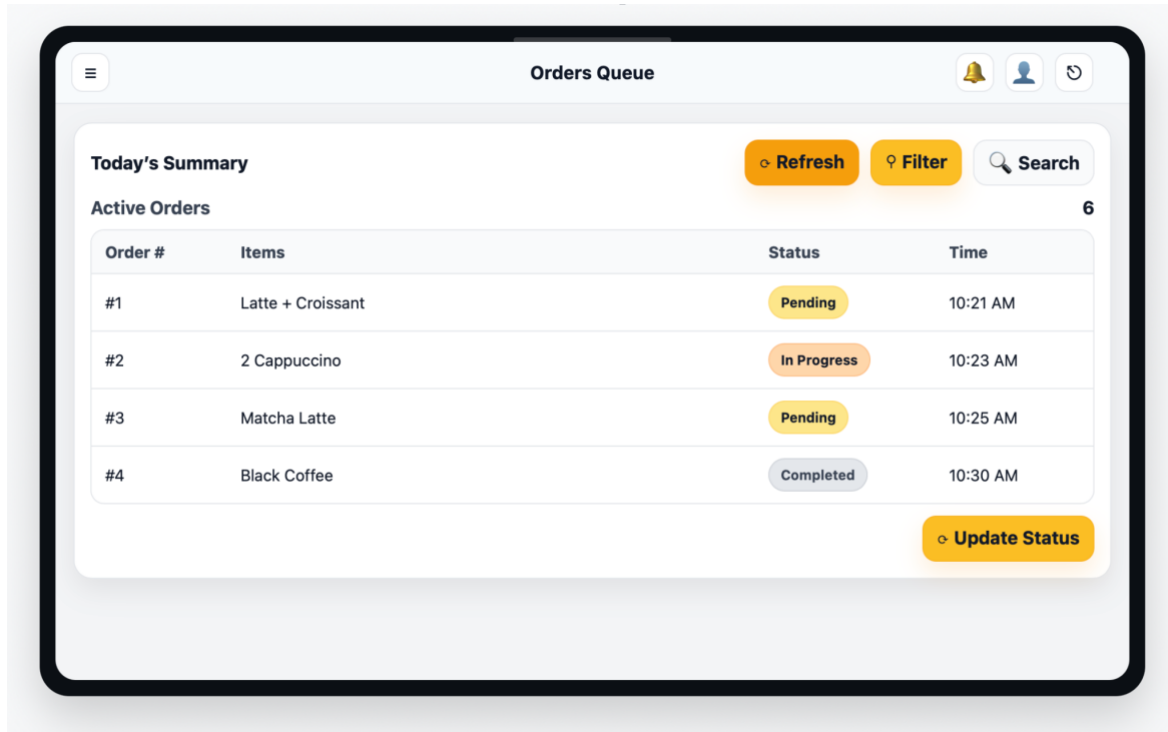
3.1 User Interfaces

The following pages provide tablet POS (wide landscape) user interface mockups for the Coffee Shop POS System. Each screen uses a consistent design language: top navigation bar, light gray background, white cards, and yellow primary action buttons.

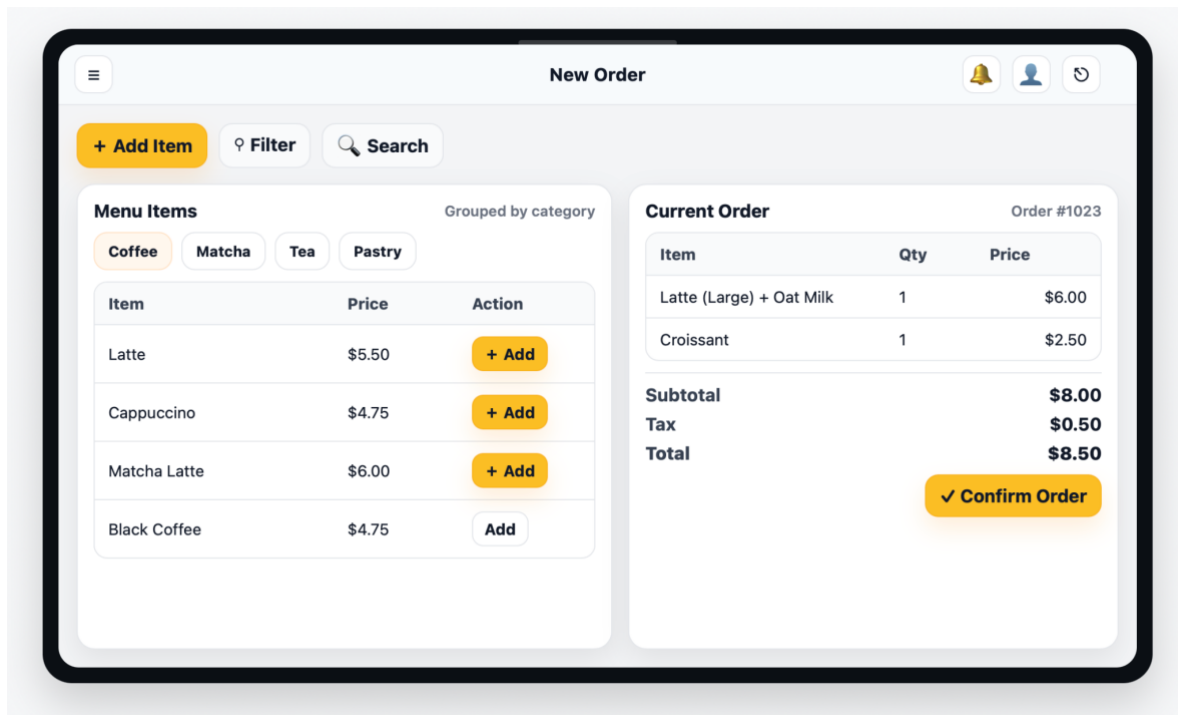
1. Barista Dashboard



2. Orders Queue



3. New Order



4. Checkout

☰

Checkout



Order Summary

Order #1023

Item	Qty	Price
Latte (Large) + Oat Milk	1	\$6.00
Croissant	1	\$2.50
Subtotal		\$8.00
Tax		\$0.50
Total		\$8.50

Payment Method

Select one

Cash

Card

Mobile Pay

Cash Received

Amount

Change Due

Example

\$10.00

\$1.50

✓ Confirm Payment

5. Admin Dashboard

☰

Admin Center



Quick Stats

Daily Weekly Monthly Custom Range

Today's Revenue

Transactions

Best Seller

\$3,420

189

Latte

Shortcuts

Admin tools



Manage Menu

Add / edit items



Inventory

Stock & thresholds



Reports

Revenue analytics



Export

CSV / PDF



Staff Accounts

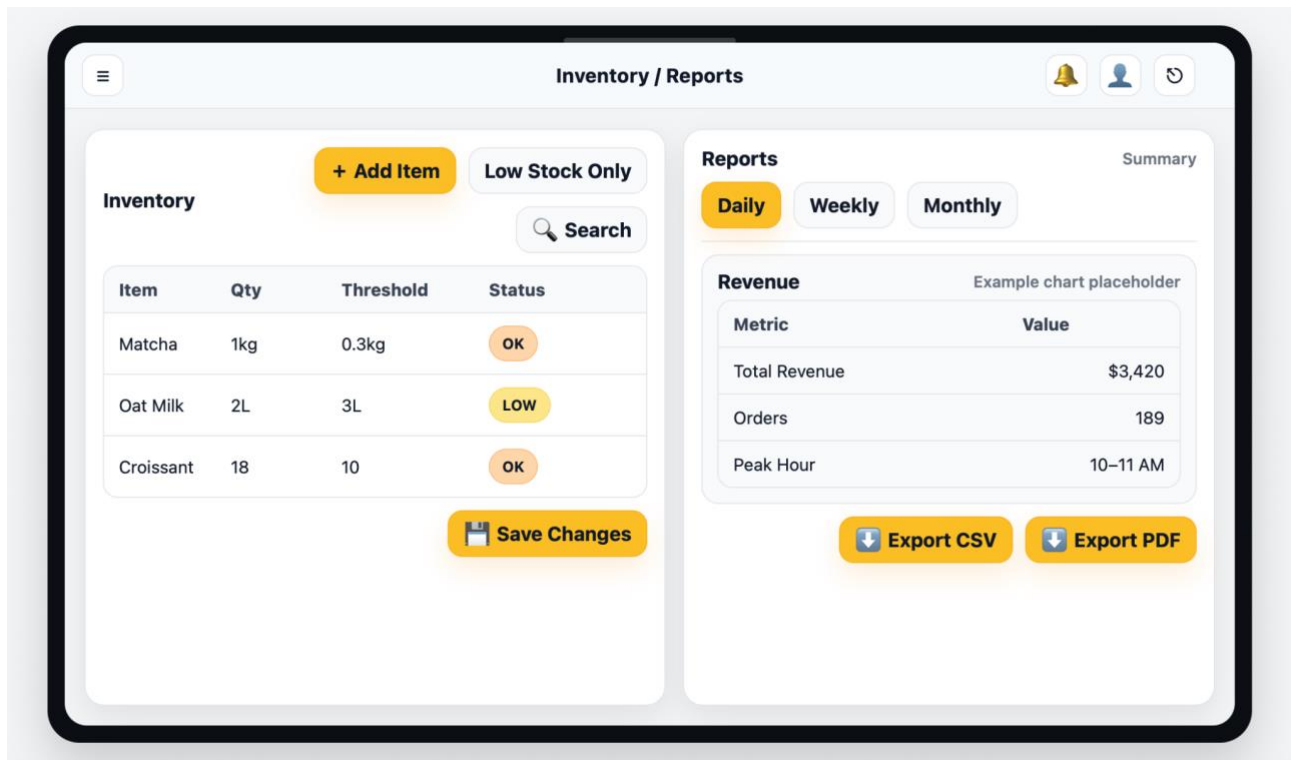
Roles / access



Refunds

Full / partial

6. Inventory Reports



3.2 Hardware Interfaces

The Coffee POS system interacts with the following hardware components used in the point-of-sale environment

- POS Client Devices

The system operates on POS client devices used by staff and administrators. These devices include tablets or desktop POS terminals with touchscreen capability. The devices provide the primary interface for order creation, payment processing, inventory management, and administrative functions.

3.3 Software Interfaces

a) Database

- **Type:** Relational SQL (PostgreSQL / MySQL compatible).

- **Purpose:** Persistent storage of users, sessions, menu items, orders, transactions, inventory, and audit logs.
- **Interface requirements:** Standard SQL connections over TLS. The system will use an ORM layer (e.g., JPA/Hibernate for Java backend) to isolate DB-specific SQL and allow migration between PostgreSQL/MySQL.
- **Availability & reliability:** Database must support ACID guarantees for transaction integrity and be configured with automated backups and point-in-time recovery.

b) Backend API

- **Type:** RESTful JSON API (HTTP/HTTPS).
- **Purpose:** Primary interface between frontend (tablet/web) and backend services for authentication, order management, inventory, reporting, and admin operations.
- **Contract highlights:**
 - All endpoints accept and return JSON.
 - Standard HTTP status codes will be used (2xx success, 4xx client errors, 5xx server errors).
 - Authentication via bearer tokens (JWT) with expiry; server validates scopes/roles per endpoint.
 - Rate limiting is applied to public endpoints to protect against abuse.
- **Versioning:** API versioning via URL to allow compatible upgrades.

c) Payment Gateway Integration

- **Type:** Vendor-specific HTTPS API (PCI-DSS compliant).
- **Purpose:** Authorize/settle card and mobile payments, refunds.
- **Interface requirements:**

- All payment interactions occur server-side through the payment gateway SDK/API.
- Cardholder data must never be stored on our servers (use tokenization or gateway-hosted fields).
- TLS 1.2+ and required PCI encryption standards are enforced.
- Support for refund API and transaction query API for reconciliation.

d) Email Service

- **Type:** SMTP or transactional-email API (e.g., SendGrid, SES) over TLS.
- **Purpose:** Password resets, notifications (receipts, admin alerts), audit notifications.
- **Requirements:** Support templating, delivery webhooks (optional), and retry logic for transient failures.

e) External Reporting / Export

- **Type:** File generation (CSV, PDF) and optional integration endpoints for downstream analytics.
- **Purpose:** Export sales and transaction data for accounting and analysis.
- **Interface requirements:** Exports must match on-screen reports and be downloadable via secure admin endpoints.

3.4 Communications Interfaces

a) Protocols & Transport

- **Primary protocol:** HTTPS (HTTP/1.1 or HTTP/2) using TLS 1.2+ for all client-server and server-server communication.
- **Internal service calls:** Prefer HTTPS / gRPC for internal microservices (if introduced), with mTLS where feasible.

b) Ports & Network

- **Frontend → Backend:** TCP port 443 (HTTPS).
- **Backend → Database:** Encrypted DB port (e.g., 5432 for PostgreSQL) accessible only from backend hosts/VPC.
- **Payment gateway:** Backend outbound HTTPS (port 443) to provider endpoints.
- **Email service:** Outbound HTTPS/SMTP (TLS).

c) Message Formats

- **REST endpoints:** JSON request/response bodies following a documented schema (OpenAPI).
- **Export files:** CSV and PDF for reports. CSV must use UTF-8 and a comma delimiter; fields quoted as necessary. PDFs must use A4 or letter layout, depending on deployment locale.

d) Security & Encryption

- **TLS:** All in-transit communication must be encrypted with TLS 1.2 or newer.
- **Authentication:** JWT bearer tokens for API access; refresh tokens for session renewal.
- **Role enforcement:** Each endpoint validates roles/scopes. Sensitive admin endpoints require both authentication and admin role.
- **Network segmentation:** Database and internal services must be in private subnets with firewall rules only allowing required IP ranges.

e) Performance & Timeouts

- **Response time targets:** Most interactive endpoints should respond within 2–5 seconds under normal load (see Nonfunctional Section).
- **Connection timeouts:** Frontend requests default timeout 30s; long-running report generation should be asynchronous with job status endpoints.

f) Synchronization & Offline

- **Real-time sync:** Active order queue and inventory changes must propagate to clients in near real-time (via WebSockets or short-polling fallback).
- **Offline considerations:** POS should gracefully handle temporary network loss:
 - Allow queueing of orders locally with later reconciliation.
 - Disable payment processing that requires gateway access; allow cash payments to be recorded locally and reconciled when online.

4. System Features

4.1 Use Case 1

Use Case #1	Login/Logout	
Actor	Staff (Barista)	
Preconditions	● Staff must have a registered account ● Staff must have a network connection ● Staff is not already logged in (for login). ● Staff is logged in (for logout).	
Flow of Control (Login)	Enter username and password	

		Validates entered credentials
	Clicks the “Login” button	
		Checks account status If valid, it creates a user session Displays the dashboard screen
Flow of Control (Logout)	Staff clicks “Logout.”	
		Displays the logout confirmation dialog
	Confirms logout	
		Terminates active session Redirects the user to the login screen
Post Conditions	- Staff is successfully logged in or logged out - A session is created or terminated	

Error Conditions	• Invalid username or password • Network connection lost • The account is deactivated • Session timeout due to inactivity	
Non-functional requirements	• Login must be completed within 3 seconds • Auto-timeout after 10 minutes of inactivity	

4.2 Use Case 2

Use Case #2	Create Order	
Actor	Staff (Barista)	
Preconditions	• Staff must be logged in • Menu items must exist • The system must have a network/database connection	
Flow Of Control	Selects “New Order”	
		Displays available menu items

	Selects one or more items	
		Adds selected item(s) to the temporary order list
	Customizes item (size, milk type, add-ons)	
		Updates item details and recalculates price
	Adds items to the order	
		Displays updated order summary
	Reviews order summary	
		Shows subtotal and total price
	Confirms order	
		Saves order in the database Sets order status to "Pending." Displays a confirmation message
Post Conditions	• The order is saved in the database • Order status is set to "Pending."	

	• The order appears in the active orders list	
Error Conditions	• Item unavailable • Invalid customization • Network/database failure	
Non-functional requirements	• Order creation must be completed within 5 seconds • The system must handle multiple simultaneous orders	

4.3 Use Case 3

Use Case #3	Modify Order	
Actor	Staff (Barista)	
Preconditions	• Staff must be logged in • Order must exist • Payment must not yet be processed	
Flow of Control	Selects an existing order from the active orders list	
		Displays order details and item breakdown

	Edits items, quantities, or customizations	
		Updates order details in the temporary view
	Removes or adds items as needed	
		Recalculates the subtotal and total price
	Clicks “Save Changes”	
		Validates modifications Displays a confirmation message Updates order information in the database
Post Conditions	- Order details are updated in the database - The updated order is displayed	
Error Conditions	- Order not found - Network connection lost	
Non-functional	Modification must be	

requirements	completed within 3 seconds	
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4.4 Use Case 4

Use Case #4	Process Payment	
Actor	Staff (Barista)	
Preconditions	• Staff must be logged in • Order must exist • The order must not be paid for	
Flow of Control	Selects “Process Payment” for an order	
		Displays available payment options (cash, credit/debit, mobile payment)
	Selects payment type	
		Prompts for payment details (if electronic) or cash amount
	Enter payment information or confirm cash received	

		Validates payment details
	Confirms payment	
		Processes transactions through the payment gateway (if electronic) Records payment in the database Updates order status to "Completed." Generates and displays a receipt Displays payment confirmation message
Post Conditions	• Payment recorded in the database • Order status updated to "Completed." • Receipt generated	
Error Conditions	• Payment declined • Insufficient funds • Network or payment gateway failure	
Non-functional requirements	• Payment processing must be completed within 5 seconds	

	Transaction data must be securely encrypted	
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4.5 Use Case 5

Use Case #5	Update Order Status	
Actor	Staff (Barista)	
Preconditions	- Staff must be logged in - Order must exist	
Flow of Control	Staff selects the order	
		Displays current order details and status
	Staff updates status (Pending -> In Progress -> Completed)	
		Validates status transition
	Confirms status update	
		- Saves updated status in the database - Refreshes order display - Updates the order in the

		active orders queue • Displays a confirmation message
Post Conditions	• Order status updated in the database • Status visible to relevant users	
Error Conditions	• Invalid status transition • Network failure	
Non-functional requirements	• Status update must be completed within 2 seconds	

4.6 Use Case 6

Use case #6	Staff Menu
Actor	Staff
Pre-condition	• Staff is logged into the system and has appropriate permissions • Allow staff to view the current menu, including prices, availability status, and operational details (item status, hidden items, and stock levels).

	The system has a stable connection	
Flow of control		
	Staff logs into the system.	
		The system authenticates credentials and grants access based on role.
	Staff selects View Menu from the dashboard	
		The system retrieves and displays the current menu list grouped by categories.
	Staff selects a menu item	
		The system displays detailed information (price, availability, stock level, status: active/inactive).
	Staff reviews availability (items marked out of stock)	
		The system shows real-time status and highlights unavailable items.

	Staff refreshes or navigates back	
		The system updates menu data if needed and returns to the previous screen.
Post-condition	Staff successfully views the current menu status. No changes are made unless another use case is triggered.	
Error- condition	- Incorrect role permissions are exposing restricted data - Inventory synchronization delays - Performance issues with a large menu	
Non-functional requirements	- The system must allow staff logged in with valid credentials. - Menu items display name, category, price, and availability with less than 3s - The menu reflects real-time updates.	

4.7 Use Case 7

Use Case #7	View Active Orders Queue	
Actor	Staff (Barista)	

Preconditions	• The staff user is logged into the POS system. • The system database is operational. • There are existing orders in the pending or in-progress states. • The user has permission to access the order queue. • The Internet/network connection is active.	
Flow of Control	The barista selects “Active Orders” from the dashboard.	
		The system queries the database for all orders with status = Pending or In Progress.
	The barista views the order queue.	
		The system displays orders in timestamp order (FIFO).
	The barista selects a specific order.	
		The system displays order

		details: Order ID, items, customizations, payment status, and timestamp.
	The barista chooses to update the order status	
		System updates the order status in the database.
	Barista marks the order as "completed."	
		System removes the order from Active Queue and logs completion time.
	Barista refreshes page.	
		System reloads and syncs queue in real-time without duplicating records.
Post Conditions	● Active order list reflects real-time database state. ● Completed orders are removed from queue.	

	• Updated status is saved permanently. • Order details are accurately displayed.	
Error Conditions	• If database fails → Display error message: “Unable to retrieve active orders.” • If no active orders → Display message: “No active orders available.” • If update fails → Display: “Order update unsuccessful. Try again.” • If network lost → Show offline indicator and disable updates. • If unauthorized access → Display access denied message.	
Non-functional requirements	• Real-time refresh must occur within 2 seconds. • The system must handle a	

	minimum 100 concurrent active orders. • The interface must be optimized for tablet POS screens. • Queues must remain responsive under peak hours. • System uptime: 99% during business hours.	
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4.8 Use Case 8

Use case #8	Owner's menu	
Actor	Owner/admin	
Pre-condition	• Admin authentication • System has stable connection	
Flow of control	Admin navigates to Menu and selects Add Item	
		• The system validates the

		user to ensure the user is authorized admin. • The system presents an item creation form with a field: items, price, description, item available.
	Admin fills out the form	
		• The system validates the input. • The system displays user's input
	Admin previews the item, click save if agree or modified if not	
		• If the admin agrees what is displayed, the system creates the items record. • The system pushes the update to live menu • If the admin does not

		agree, the system allows the admin to modify the data.
	Admin receives the confirmation after reviewing	
		System displays success message and ask if the admin want to log out
	Admin log out	
		System log out successfully
Post-condition	Items are created/updated/deactivated in the menu database; audit trail recorded; live menu reflects changes once published/synced	
Error- condition	● If required fields are missing or invalid (e.g., negative price), system rejects save and displays field-specific errors. User corrects and retries. ● If user lacks permission, system denies access and logs attempt. ● If another admin edited before saving, system warns and shows different admin can merge, discard, or force overwrite.	

	• If publish to downstream systems fails, system saves locally, marks update as pending and retries
Non-functional requirements	• Item create/update actions should complete within 2 seconds for single items. • All actions require strong authentication • Keep immutable audit logs

4.9 Use Case 9

Use Case #9	Manage Inventory	
Actors	Admin (Owner/Manager)	
Pre-Conditions	• Admin must be logged in • Inventory items must exist in database • Network/Database connection must be available	
Flow of Control	Admin opens “Inventory” dashboard	
		System displays current stock levels + low-stock thresholds
	Admin updates quantities or thresholds	
		System validates values (non-negative, reasonable bounds)

	Admin presses “Save”	
		System commits updates and syncs changes in real time
Post Conditions	• Inventory levels/thresholds updated and stored • Updated values affect order availability rules	
Error Condition	• Invalid quantity/threshold • Inventory record not found • Network/Database failure to connect	
Non-Functional Requirements	• Inventory updates visible across system in real time • Audit trail recorded for inventory changes (recommended for integrity)	

4.10 Use Case 10

Use Case #10	View Sales Report	
Actor	Admin (Owner/Manager)	
Pre-Conditions	• Admin must be logged in • Sales/Transaction data exists • Network/Database connection is available	
Flow of Control	Admin opens “Reports” dashboard	

		System shows report types (Daily/Weekly/Monthly, revenue, best-sellers, peak times, etc.)
	Admin selects date range/report type	
		System aggregates sales metrics and displays charts/tables
	Admin drills into details (selects a specific metric, date, or product)	
		System displays detailed transaction list and item-level breakdown
Post Conditions	Admin sees requested report and metrics details	
Error Condition	- No data for selected range - Report generation failure - Network/Database failure to connect	
Non-Functional Requirements	- Reports load within 5 seconds for typical date ranges - Access restricted to Admin role (no staff access)	

4.11 Use Case 11

Use Case #11	Export Reports	
Actor	Admin (Owner/Manager)	
Pre-Conditions	• Admin must be logged in • A report is generated/available to export	
Flow of Control	Admin selects “Export” on a report	
		System shows export formats (CSV/PDF)
	Admin chooses format	
		System generates file
	Admin downloads file	
		System provides downloadable file and confirms export success
Post Conditions	• Export file generated and delivered to admin in (CSV/PDF) format	
Error Condition	• Export generation fails • Network error during download • Network/Database failure to connect	

Non-Functional Requirements	• Export completes within 10 seconds for typical reports • Exported files match on-screen data (consistency)
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4.12 Use Case 12

Use Case #12	Manage Staff Accounts	
Actor	Admin (Owner/Manager)	
Pre-Conditions	• Admin must be logged in • Admin has user-management permissions • Network/Database connection is available	
Flow of Control	Admin opens “Staff Accounts”	
		System displays list of staff users and roles
	Admin selects Create, Edit, or Deactivate	
		System displays account form
	Admin enters or modifies user information	

		System validates input and role assignments
	Admin clicks “Save”	
		• System updates user record • System logs action • System displays confirmation message
Post Conditions	• Staff account created, updated, or deactivated • Role permissions applied immediately • Audit trail recorded	
Error Condition	• Duplicate username/email (if enforced) • Invalid role assignment • Network/Database failure to connect	
Non-Functional Requirements	• RBAC must be enforced at all times • Authentication and account	

	changes must be logged • Changes must complete within 3 seconds	
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4.13 Use Case 13

Use Case #13	Process Refund	
Actor	Admin (Owner/Manager)	
Pre-Conditions	• Admin must be logged in • Transaction exists and is eligible for refund • Network/Database/Payment system connection is available (E-Refunds)	
Flow of Control	Admin opens “Transaction History”	
		System displays searchable transaction list
	Admin selects a completed transaction	
		System displays transaction details

		and refund option
	Admin selects refund type (Full/Partial)	
		System validates refund eligibility
	Admin confirms refund	
		• System processes refund through payment gateway • System updates transaction record • System logs refund action • System displays confirmation message
Post Conditions	• Refund recorded with timestamp • Transaction marked Refunded or Partially Refunded • Audit record stored	
Error Condition	• Transaction not found • Refund not permitted	

	• Payment gateway failure • Network/Database failure	
Non-Functional Requirements	• Refund processing must be secure and encrypted • Refund actions must be logged • Refund must complete within 5 seconds	

4.14 Use Case 14

Use Case #14	View Transaction Logs	
Actor	Admin (Owner/Manager)	
Pre-Conditions	• Admin must be logged in • Transaction log data exists • Network/Database connection is available	
Flow of Control	Admin opens “Transaction Logs”	
		System displays searchable log list (date, amount, method, order reference)

	Admin filters or searches logs	
		System refreshes list with matching results
	Admin selects a transaction	
		System displays full transaction details
Post Conditions	Admin can audit transactions and view details	
Error Condition	- No results found for selected filter - Network/Database failure - Unauthorized access attempt	
Non-Functional Requirements	- Logs must load within 3–5 seconds - Access restricted to Admin role - Log data must be tamper-resistant	

5. Nonfunctional Requirements

5.1 Performance Requirements

1. Login Response Time

The system shall authenticate staff login credentials and display the dashboard within **3 seconds** under normal network conditions.

Source Use Case: Login/Logout

2. Order Creation Speed

The system shall process a new order and display it in the order queue within **2 seconds** after confirmation.

Source Use Case: Create Order

3. Order Status Update Speed

The system shall update order statuses (e.g., Pending → Preparing → Completed) within **2 seconds** of staff input.

Source Use Case: Update Order Status

4. Checkout Processing Time

The system shall complete payment authorization and transaction confirmation within **3 seconds** for standard payment methods.

Source Use Case: Checkout / Process Payment

5. Menu Management Response Time

Menu item creation, update, or deactivation actions shall complete within **2 seconds** for individual items.

Source Use Case: Manage Menu

6. Transaction Log Retrieval

The system shall load filtered transaction logs within **3–5 seconds** for typical search queries.

Source Use Case: View Transaction Logs

5.2 Usability Requirements

1. Touchscreen Optimization

The POS interface shall be optimized for touchscreen devices with clearly labeled buttons and minimal navigation steps.

2. NFR-U2: Intuitive Workflow

Staff shall be able to create and complete an order in **three steps or fewer** from the dashboard.

3. Training Time

New staff members shall be able to learn the basic POS functions within **30 minutes of training**.

4. Clear Error Feedback

The system shall display clear error messages and guidance if an invalid input or payment failure occurs.

Source Use Cases: Create Order, Checkout, Update Order Status

5.3 Reliability and Capacity Requirements

1. Concurrent Order Handling

The system shall support **at least 100 concurrent active orders** without degradation of performance.

2. Peak Hour Responsiveness

The system shall remain responsive and allow order processing during peak business hours without crashes or queue failures.

3. Simultaneous Order Creation

The system shall handle multiple simultaneous order creations by different staff members without data conflicts.

4. System Availability

The system shall maintain **99.5% uptime** during business hours.

Source Use Cases: Create Order, View Active Orders Queue

5.4 Security Requirements

1. Authentication Requirement

Staff must authenticate with a valid username and password before accessing the POS system.

2. Role-Based Access Control

Administrative features (menu management, reports, transaction logs) shall only be accessible to authorized administrators.

3. Session Timeout

User sessions shall automatically log out after **10 minutes of inactivity**.

4. Data Encryption

Sensitive information such as payment details and login credentials shall be transmitted using **secure encrypted communication (HTTPS/TLS)**.

Source Use Cases: Login/Logout, Manage Menu, View Reports

5.5 Data Integrity and Logging Requirements

1. Transaction Logging

All completed transactions shall be recorded in a secure database with timestamps and order details.

2. Immutable Audit Logs

Audit logs shall be **append-only and tamper-resistant** to prevent unauthorized modification.

3. Administrative Action Logging

Administrative actions such as menu edits, price changes, and inventory updates shall be recorded in system logs.

4. Consistent Report Generation

Reports generated by the system shall accurately reflect stored transaction data and allow export in **CSV or PDF format**.

Source Use Cases: Manage Menu, View Reports, View Transaction Logs

6. Other Requirements

a) Maintainability

- Codebase must follow the project's style guide and include automated unit and integration tests.
- CI/CD pipeline required for automated builds, tests, and deployments.
- All public APIs must be covered by API documentation (OpenAPI) and kept in sync with releases.

b) Scalability

- The system should scale horizontally for stateless backend components (API servers) behind a load balancer.
- Database should be provisioned with read replicas for reporting and scaling read-heavy workloads.
- Design must support at least a stated target throughput (e.g., 100 concurrent active orders per store during peak hours; adjust per deployment)